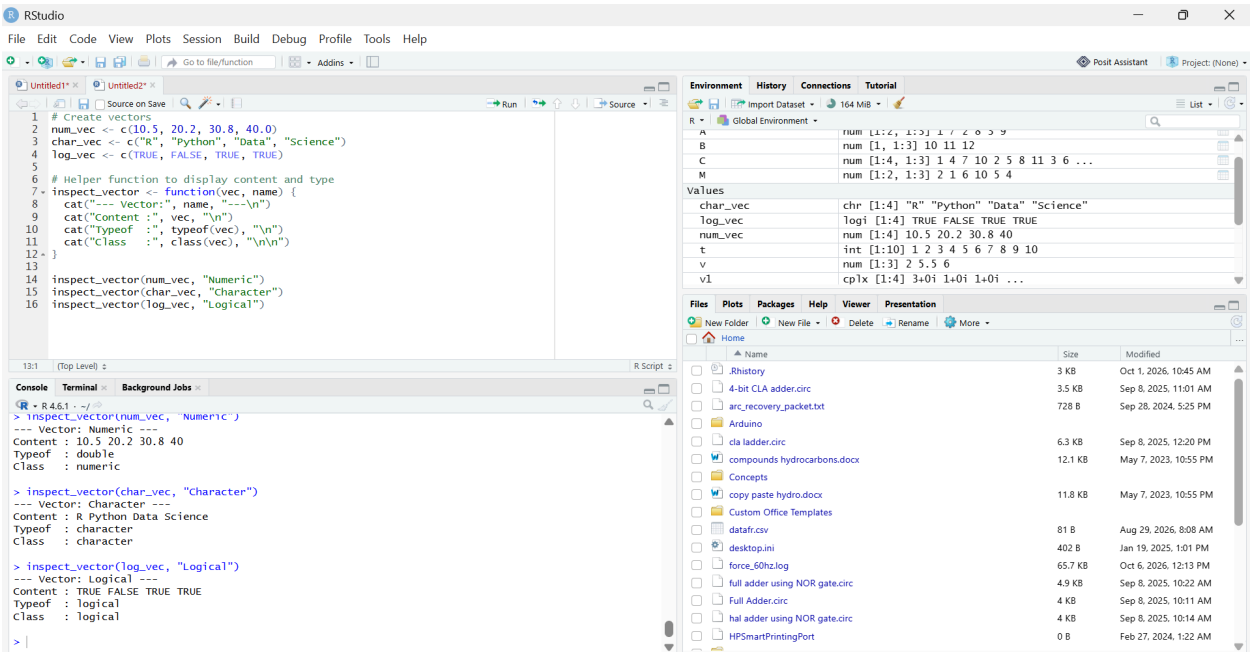
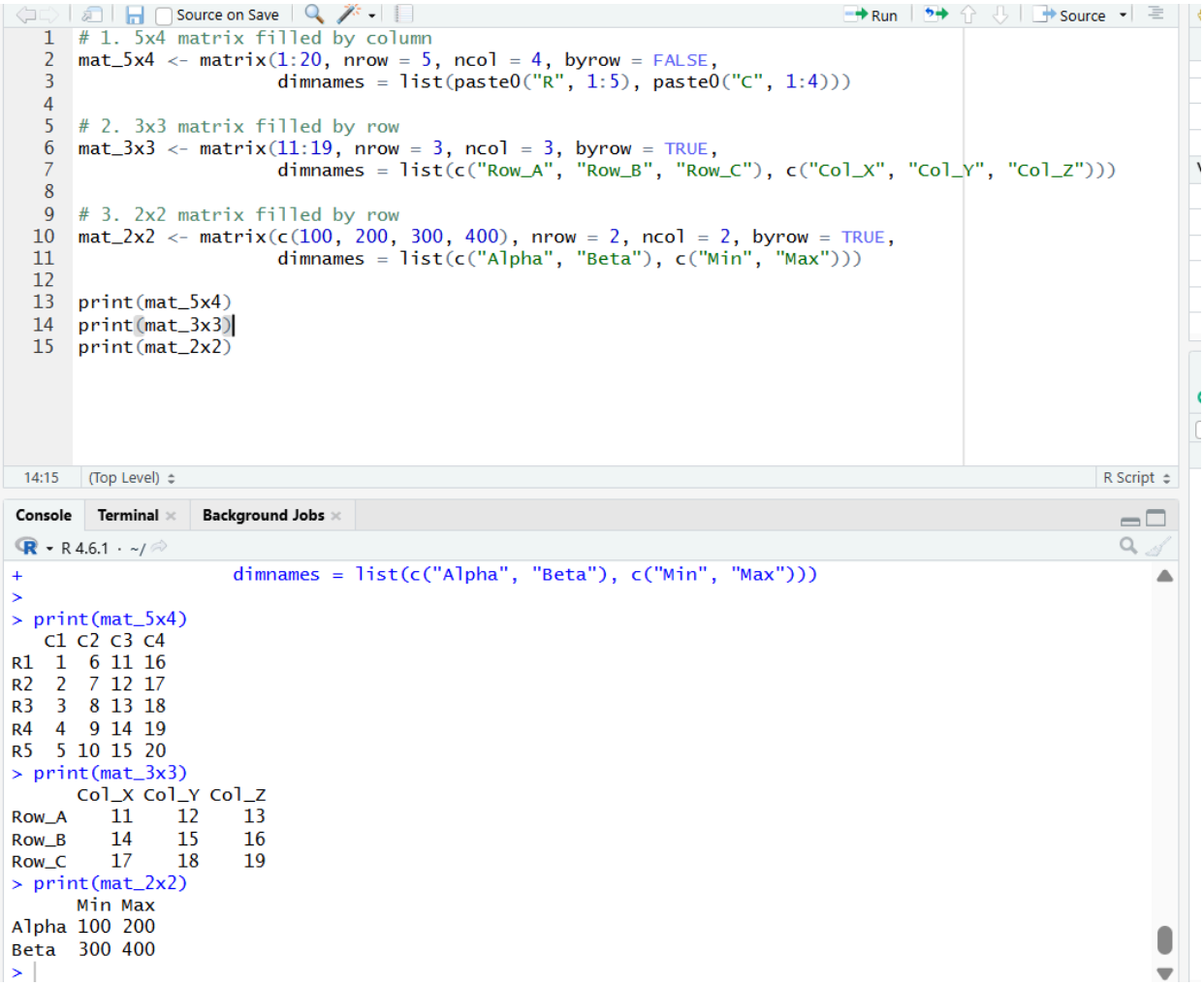
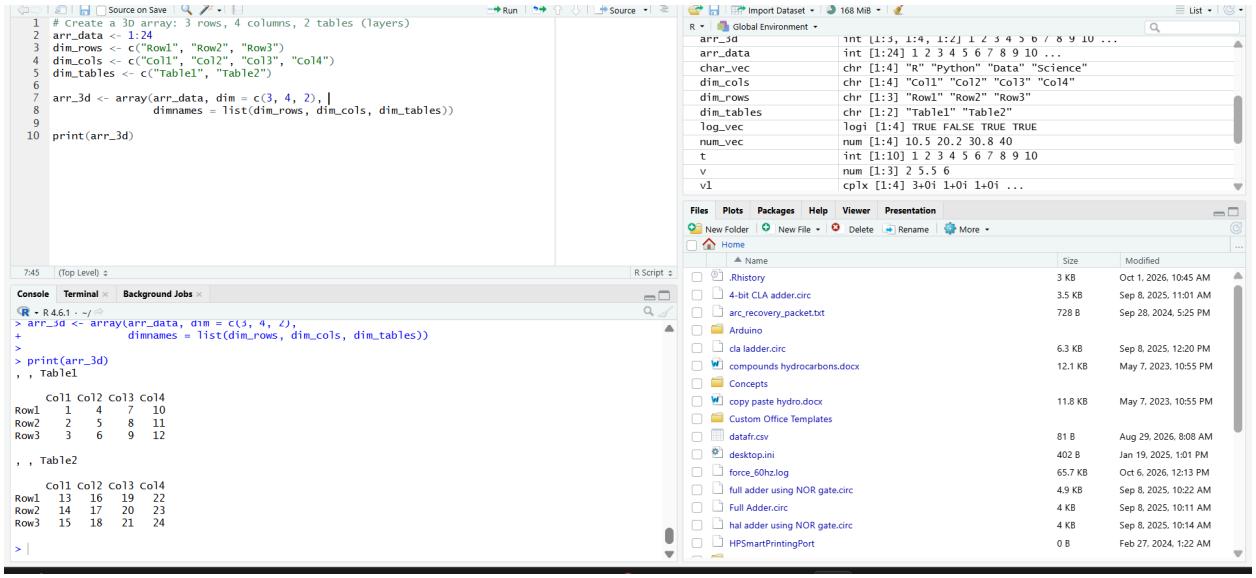
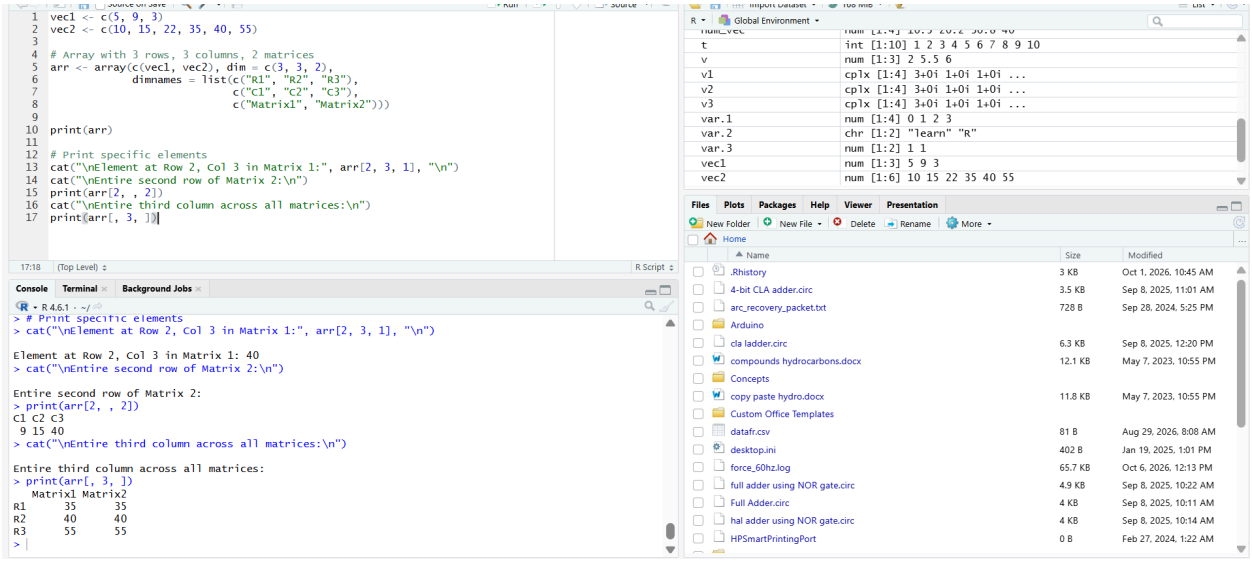

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Untitled2*
Create vectors
num_vec <- c(10.5, 20.2, 30.8, 40.0)
char_vec <- c("R", "Python", "Data", "Science")
log_vec <- c(TRUE, FALSE, TRUE, TRUE)
Helper function to display content and type
inspect_vector <- function(vec, name) {
cat("--- Vector:", name, "---\n")
cat("Content :", vec, "\n")
cat("Typeof :", typeof(vec), "\n")
cat("Class :", class(vec), "\n\n")
}
inspect_vector(num_vec, "Numeric")
inspect_vector(char_vec, "Character")
inspect_vector(log_vec, "Logical")
Console
Terminal
Background Jobs
--- Vector: Numeric ---
Content : 10.5 20.2 30.8 40
Typeof : double
Class : numeric
--- Vector: Character ---
Content : R Python Data Science
Typeof : character
Class : character
--- Vector: Logical ---
Content : TRUE FALSE TRUE TRUE
Typeof : logical
Class : logical
Environment
History
Connections
Tutorial
B num [1, 1:3] 10 11 12
C num [1:4, 1:3] 1 4 7 10 2 5 8 11 3 6 ...
M num [1:2, 1:3] 2 1 6 10 5 4
Values
char_vec chr [1:4] "R" "Python" "Data" "Science"
log_vec logi [1:4] TRUE FALSE TRUE TRUE
num_vec num [1:4] 10.5 20.2 30.8 40
t int [1:10] 1 2 3 4 5 6 7 8 9 10
v num [1:3] 2 5.5 6
v1 cplx [1:4] 3+0i 1+0i 1+0i ...
Files
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4-bit CLA adder.circ
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full adder using NOR gate.circ
Full Adder.circ
hal adder using NOR gate.circ
HPSmartPrintingPort


Source on Save
Run
Source
1. 5x4 matrix filled by column
mat_5x4 <- matrix(1:20, nrow = 5, ncol = 4, byrow = FALSE,
dimnames = list(paste0("R", 1:5), paste0("C", 1:4)))
2. 3x3 matrix filled by row
mat_3x3 <- matrix(11:19, nrow = 3, ncol = 3, byrow = TRUE,
dimnames = list(c("Row_A", "Row_B", "Row_C"), c("Col_X", "Col_Y", "Col_Z")))
3. 2x2 matrix filled by row
mat_2x2 <- matrix(c(100, 200, 300, 400), nrow = 2, ncol = 2, byrow = TRUE,
dimnames = list(c("Alpha", "Beta"), c("Min", "Max")))
print(mat_5x4)
print(mat_3x3)
print(mat_2x2)
14:15
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
+ dimnames = list(c("Alpha", "Beta"), c("Min", "Max")))
>
> print(mat_5x4)
C1 C2 C3 C4
R1 1 6 11 16
R2 2 7 12 17
R3 3 8 13 18
R4 4 9 14 19
R5 5 10 15 20
> print(mat_3x3)
Col_X Col_Y Col_Z
Row_A 11 12 13
Row_B 14 15 16
Row_C 17 18 19
> print(mat_2x2)
Min Max
Alpha 100 200
Beta 300 400
>


Source on Save
Run
Source
Create a 3D array: 3 rows, 4 columns, 2 tables (layers)
arr_data <- 1:24
dim_rows <- c("Row1", "Row2", "Row3")
dim_cols <- c("Col1", "Col2", "Col3", "Col4")
dim_tables <- c("Table1", "Table2")
arr_3d <- array(arr_data, dim = c(3, 4, 2),
dimnames = list(dim_rows, dim_cols, dim_tables))
print(arr_3d)
7:45
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.4.1 · ~/
> arr_3d <- array(arr_data, dim = c(3, 4, 2),
+ dimnames = list(dim_rows, dim_cols, dim_tables))
>
> print(arr_3d)
, , Table1
Col1 Col2 Col3 Col4
Row1 1 4 7 10
Row2 2 5 8 11
Row3 3 6 9 12
, , Table2
Col1 Col2 Col3 Col4
Row1 13 16 19 22
Row2 14 17 20 23
Row3 15 18 21 24
>
Import Dataset
168 MiB
List
R
Global Environment
arr_3d int [1:3, 1:4, 1:2] 1 2 3 4 5 6 7 8 9 10 ...
arr_data int [1:24] 1 2 3 4 5 6 7 8 9 10 ...
char_vec chr [1:4] "R" "Python" "Data" "Science"
dim_cols chr [1:4] "Col1" "Col2" "Col3" "Col4"
dim_rows chr [1:3] "Row1" "Row2" "Row3"
dim_tables chr [1:2] "Table1" "Table2"
log_vec logi [1:4] TRUE FALSE TRUE TRUE
num_vec num [1:4] 10.5 20.2 30.8 40
t int [1:10] 1 2 3 4 5 6 7 8 9 10
v num [1:3] 2 5.5 6
v1 cplx [1:4] 3+0i 1+0i 1+0i ...
Files
Plots
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Help
Viewer
Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
.Rhistory 3 KB Oct 1, 2026, 10:45 AM
4-bit CLA adder.circ 3.5 KB Sep 8, 2025, 11:01 AM
arc_recovery_packet.txt 728 B Sep 28, 2024, 5:25 PM
Arduino
cla ladder.circ 6.3 KB Sep 8, 2025, 12:20 PM
compounds hydrocarbons.docx 12.1 KB May 7, 2023, 10:55 PM
Concepts
copy paste hydro.docx 11.8 KB May 7, 2023, 10:55 PM
Custom Office Templates
datafr.csv 81 B Aug 29, 2026, 8:08 AM
desktop.ini 402 B Jan 19, 2025, 1:01 PM
force_60hz.log 65.7 KB Oct 6, 2026, 12:13 PM
full adder using NOR gate.circ 4.9 KB Sep 8, 2025, 10:22 AM
Full Adder.circ 4 KB Sep 8, 2025, 10:11 AM
hal adder using NOR gate.circ 4 KB Sep 8, 2025, 10:14 AM
HPSmartPrintingPort 0 B Feb 27, 2024, 1:22 AM



```
vec1 <- c(5, 9, 3)
vec2 <- c(10, 15, 22, 35, 40, 55)

# Array with 3 rows, 3 columns, 2 matrices
arr <- array(c(vec1, vec2), dim = c(3, 3, 2),
             dimnames = list(c("R1", "R2", "R3"),
                             c("C1", "C2", "C3"),
                             c("Matrix1", "Matrix2")))

print(arr)

# Print specific elements
cat("\nElement at Row 2, Col 3 in Matrix 1:", arr[2, 3, 1], "\n")
cat("\nEntire second row of Matrix 2:\n")
print(arr[2, , 2])
cat("\nEntire third column across all matrices:\n")
print(arr[, 3, ])
```

```
> # Print specific elements
> cat("\nElement at Row 2, Col 3 in Matrix 1:", arr[2, 3, 1], "\n")

Element at Row 2, Col 3 in Matrix 1: 40
> cat("\nEntire second row of Matrix 2:\n")

Entire second row of Matrix 2:
> print(arr[2, , 2])
C1 C2 C3
 9 15 40
> cat("\nEntire third column across all matrices:\n")

Entire third column across all matrices:
> print(arr[, 3, ])
   Matrix1 Matrix2
R1      35      35
R2      40      40
R3      55      55
>
```

t	int [1:10] 1 2 3 4 5 6 7 8 9 10
v	num [1:3] 2 5.5 6
v1	cplx [1:4] 3+0i 1+0i 1+0i ...
v2	cplx [1:4] 3+0i 1+0i 1+0i ...
v3	cplx [1:4] 3+0i 1+0i 1+0i ...
var.1	num [1:4] 0 1 2 3
var.2	chr [1:2] "learn" "R"
var.3	num [1:2] 1 1
vec1	num [1:3] 5 9 3
vec2	num [1:6] 10 15 22 35 40 55

Name	Size	Modified
.Rhistory	3 KB	Oct 1, 2026, 10:45 AM
4-bit CLA adder.circ	3.5 KB	Sep 8, 2025, 11:01 AM
arc_recovery_packet.txt	728 B	Sep 28, 2024, 5:25 PM
Arduino		
cla ladder.circ	6.3 KB	Sep 8, 2025, 12:20 PM
compounds hydrocarbons.docx	12.1 KB	May 7, 2023, 10:55 PM
Concepts		
copy paste hydro.docx	11.8 KB	May 7, 2023, 10:55 PM
Custom Office Templates		
datafr.csv	81 B	Aug 29, 2026, 8:08 AM
desktop.ini	402 B	Jan 19, 2025, 1:01 PM
force_60hz.log	65.7 KB	Oct 6, 2026, 12:13 PM
full adder using NOR gate.circ	4.9 KB	Sep 8, 2025, 10:22 AM
Full Adder.circ	4 KB	Sep 8, 2025, 10:11 AM
hal adder using NOR gate.circ	4 KB	Sep 8, 2025, 10:14 AM
HPSmartPrintingPort	0 B	Feb 27, 2024, 1:22 AM

```
# Example A: Factorizing continuous data from the built-in 'women' dataset
data(women)
height_categories <- cut(women$height, breaks = c(55, 62, 68, 75),
                         labels = c("Short", "Medium", "Tall"))
print(table(height_categories))

# Example B: Sample of random LETTERS
set.seed(42)
letter_sample <- sample(LETTERS[1:5], size = 15, replace = TRUE)
factor_letters <- factor(letter_sample)

cat("Original factor:\n")
print(factor_letters)
cat("Factor levels:", levels(factor_letters), "\n")

# Manipulate levels: rename 'A' and 'B'
levels(factor_letters)[levels(factor_letters) %in% c("A", "B")] <- "Group_AB"
cat("\nFactor after merging levels:\n")
print(factor_letters)
```

```
>
> cat("Original factor:\n")
Original factor:
> print(factor_letters)
 [1] A E A A B D B B A D A E D B B
Levels: A B D E
> cat("Factor levels:", levels(factor_letters), "\n")
Factor levels: A B D E
>
> # Manipulate levels: rename 'A' and 'B'
> levels(factor_letters)[levels(factor_letters) %in% c("A", "B")] <- "Group_AB"
> cat("\nFactor after merging levels:\n")

Factor after merging levels:
> print(factor_letters)
 [1] Group_AB E        Group_AB Group_AB Group_AB D        Group_AB Group_AB Group_AB D
[11] Group_AB E        D        Group_AB Group_AB
Levels: Group_AB D E
>
```

women	15 obs. of 2 variables
Values	
arr	num [1:3, 1:3, 1:2] 5 9 3 10 15 22 35 40 55 5 ...
arr_3d	int [1:3, 1:4, 1:2] 1 2 3 4 5 6 7 8 9 10 ...
arr_data	int [1:24] 1 2 3 4 5 6 7 8 9 10 ...
char_vec	chr [1:4] "R" "Python" "Data" "Science"
dim_cols	chr [1:4] "Col1" "Col2" "Col3" "Col4"
dim_rows	chr [1:3] "Row1" "Row2" "Row3"
dim_tables	chr [1:2] "Table1" "Table2"
factor_letters	Factor w/ 3 levels "Group_AB","D",..: 1 3 1 1 1 1 2 1 1 1 2 ...

Name	Size	Modified
.Rhistory	3 KB	Oct 1, 2026, 10:45 AM
4-bit CLA adder.circ	3.5 KB	Sep 8, 2025, 11:01 AM
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HPSmartPrintingPort	0 B	Feb 27, 2024, 1:22 AM